

# Victor Solomon

Ph.D. Candidate | Georgia State University | Atlanta, GA

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## EDUCATION

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| Present<br>Aug 2023  | <b>Ph.D. in Computer Science, Georgia State University, Atlanta, GA</b> <ul style="list-style-type: none"><li>● Focus : Multimodal Machine Learning, Representation Learning, Neuroimaging, and Scientific AI</li><li>● Advisor : <a href="#">Dr. Jingyu Liu</a></li></ul> |
| Feb 2021<br>Sep 2019 | <b>M.Sc. in Computer Science, African University of Science and Technology, Abuja, Nigeria</b> <ul style="list-style-type: none"><li>● Focus : Machine Learning, Deep Learning, and Recommender Systems</li></ul>                                                          |
| Feb 2018<br>Jan 2014 | <b>B.Sc. in Computer Science, Kaduna State University, Kaduna, Nigeria</b> <ul style="list-style-type: none"><li>● Focus : Computer Science, Algorithms, Programming, and Data Analysis</li></ul>                                                                          |

## RESEARCH EXPERIENCE

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| Present<br>2025      | <b>Multimodal Machine Learning for Neuroimaging</b><br><i>Center for Translational Research in Neuroimaging and Data Science (TReNDS), Georgia State University</i> <ul style="list-style-type: none"><li>● Develop multimodal machine learning methods for jointly modeling complementary information from structural and functional neuroimaging data.</li><li>● Investigate representation learning and statistical fusion under incomplete multimodal observations, heterogeneous subject overlap, and high dimensional biomedical data.</li><li>● Developed <b>MSR-IVA</b>, a masked structural residual Independent Vector Analysis framework for state-aware fusion of structural MRI and dynamic functional network connectivity.</li><li>● Applied multimodal fusion methods to Alzheimer's disease neuroimaging data and related brain connectivity analyses.</li></ul> <div>Multimodal Learning   Neuroimaging   Representation Learning   Independent Vector Analysis</div> |
| 2025<br>2024         | <b>Time Series Decomposition for Solar Flare Forecasting</b><br><i>Data Mining Lab, Georgia State University</i> <ul style="list-style-type: none"><li>● Developed a signal decomposition framework using wavelet and Fourier transforms to improve representation of complex solar time series.</li><li>● Evaluated the resulting representations for solar flare forecasting and published the work at the 38th International FLAIRS Conference.</li></ul> <div>Time Series   Signal Processing   Scientific AI</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Feb 2021<br>Sep 2019 | <b>Multimodal Fusion for Accurate and Robust Recommender Systems under Noisy Visual and Textual Content</b><br><i>M.Sc. Research Project, African University of Science and Technology</i> <ul style="list-style-type: none"><li>● Investigated multimodal recommendation using collaborative user-item interactions together with visual and textual product information.</li><li>● Studied fusion strategies designed to remain robust when image and text modalities contain noisy, inconsistent, or weakly informative content.</li><li>● Evaluated multimodal learning approaches for combining complementary product representations in recommender systems.</li></ul> <div>Recommender Systems   Multimodal Fusion   Deep Learning</div>                                                                                                                                                                                                                                         |

## PUBLICATIONS

- 2026** Victor Solomon, Zening Fu, Rafal Angryk, Vince D. Calhoun, and Jingyu Liu, "MSR-IVA: Masked Structural Residual Independent Vector Analysis for State-Aware Fusion of Structural MRI and Dynamic Functional Network Connectivity," in *IEEE International Workshop on Machine Learning for Signal Processing (MLSP)*, Atlanta, GA, 2026. [**Accepted**] [[arXiv](#)]
- 2025** Victor Solomon, Junzhi Wen, Rafal Angryk, Manya Rampuria, Omkar Rayala, and Abdul Afrid, "Time Series Decomposition Using Wavelet and Fourier Transforms for Enhanced Solar Flare Forecasting," *The International FLAIRS Conference Proceedings*, Vol. 38, 2025. [[Paper](#)]

## PROFESSIONAL EXPERIENCE

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| Jul 2023<br>May 2022 | <b>Data Analyst</b> at <a href="#">United Bank for Africa</a> <ul style="list-style-type: none"><li>● Built data pipelines for risk modeling, performance tracking, and financial analysis using Python and SQL.</li><li>● Developed analytical workflows and dashboards to support data driven decision making.</li></ul> <div>PythonSQLData Analysis</div> |
| Aug 2021<br>Feb 2021 | <b>QA Engineer</b> at <a href="#">Field Technologies and Development Partners</a> <ul style="list-style-type: none"><li>● Designed and implemented automated test scripts using Python and Selenium.</li><li>● Developed test cases for REST API validation and user interface workflows.</li></ul> <div>PythonSeleniumREST API</div>                        |
| Apr 2019<br>Apr 2018 | <b>Teaching Assistant</b> at <a href="#">Plateau State University</a> <ul style="list-style-type: none"><li>● Supported undergraduate instruction, tutorials, assignment preparation, and grading for <i>Introduction to Algorithms and Data Structures</i>.</li></ul> <div>TeachingAlgorithmsMentoring</div>                                                |

## HONORS AND AWARDS

2025 Martin D. Fraser Graduate Student Conference Travel Award : Georgia State University  
2019 AUST Merit Scholarship : African University of Science and Technology  
2017 World Bank Pan-African Undergraduate Scholarship

## TECHNICAL SKILLS

|                    |                                                                                                                                                                        |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming</b> | Python, C/C++, R, Java, SQL                                                                                                                                            |
| <b>Tools</b>       | PyTorch, TensorFlow, Linux, Git, Apache Kafka, REST APIs, Selenium                                                                                                     |
| <b>Research</b>    | Multimodal Learning, Representation Learning, Deep Learning, Graph Learning, Neuroimaging, Recommender Systems, Scientific AI, Time Series Modeling, Signal Processing |